VALIDATION PLAN

OCTOBER 2026

AI LIVE DIRECTOR

An automated vision mixer that directs live multi-camera broadcasts on its own.



THE PROBLEM

EVERY LIVE CUT STILL DEPENDS ON ONE PERSON

01

MANY FEEDS, ONE PAIR OF EYES

A director has to watch every camera at once and pick the next shot in real time.

02

QUALITY RIDES ON ATTENTION

A late cut, a blurry feed or a missed reaction goes straight to air.

03

SMALL PRODUCTIONS GO WITHOUT

Events that cannot staff a control room fall back to a single fixed camera.

Multi-camera coverage stays hard to staff, and each broadcast rides on a single seat.

THE IDEA

A VIRTUAL DIRECTOR THAT WATCHES EVERY CAMERA AND MAKES EVERY CUT.

It takes in multiple camera feeds, analyzes them in real time and switches to the best angle automatically, with no one at the mixer.

HOW IT WORKS

FOUR STAGES TURN RAW FEEDS INTO A FINISHED PROGRAM

01 · INGEST

FEEDS IN

Every camera arrives at once, with near-zero latency.

NDI · RTSP · SRT

02 · ANALYZE

AI VISION

Color, action and faces are read on every feed.

YOLOv8 · MediaPipe · DeepFace

03 · DECIDE

DIRECTOR LOGIC

A rule-based state machine picks the next shot.

Broadcast rules

04 · SWITCH

CLEAN CUT

A command tells the switcher which camera to take.

OBS Studio · vMix

The whole loop, from reading a frame to commanding the switcher, is built to run in under 100 ms.

WHAT THE AI SEES

THE AI READS COLOR, ACTION AND FACES ON EVERY FEED

COLOR CALIBRATION

Matches every feed to a master camera, so cuts never jump in color.

`Histogram matching · dynamic LUTs`

ACTION TRACKING

Follows the ball and the players, and reads body pose to find the play.

`YOLOv8 · MediaPipe`

VIP PRIORITY

Recognizes celebrities and key figures, and marks their camera high priority.

`DeepFace`

Analysis runs on downscaled 480p/720p copies; full-resolution feeds are kept for the output.

THE DIRECTOR LOGIC

FOUR BROADCAST RULES KEEP EVERY CUT CLEAN

1

WIDE DURING HIGH ACTION

Stay on wide angles while play is fast, such as a basketball fast break.

2

REACTIONS WHEN PLAY STOPS

Cut to close-ups, the crowd or a VIP as soon as the action pauses.

3

HOLD EVERY SHOT

No cut faster than 3–4 seconds, so the picture never flickers.

4

REJECT BAD FEEDS

Blurry, shaking or out-of-focus cameras are dropped automatically.

The rules are coded as a state machine, so cuts are deliberate and never jarring.

STORYTELLING · PROPOSED

A STACK OF SMALL MODELS GIVES THE DIRECTOR A SENSE OF STORY

DETECT EVENTS

E2E-SPOT · T-DEED

Spots goals, fouls and missed shots with a timestamp.

FOLLOW PLAYERS

BYTETRACK · BOT-SORT

Tracks each player over time on top of YOLOv8.

IDENTIFY PLAYERS

JERSEY-NUMBER READER

Tells the system who the moment happened to.

LABEL EACH CAMERA

SHOT CLASSIFIER

Marks every feed as wide, close-up, bench or crowd.

HEAR THE GAME

AUDIO CLASSIFIER

Picks up the whistle and the crowd roar.

CHOOSE THE SHOT

SEQUENCE MODEL

Starts as written rules, then learns from human cuts.

THE OUTPUT

THE AI COMMANDS THE SWITCHER INSTEAD OF REPLACING IT

PYTHON SCRIPT

AI DIRECTOR

Decides which camera should be live.

COMMAND



OBS STUDIO · VMIX

SWITCHER

Executes the clean cut on command.

CLEAN CUT



FULL RESOLUTION

PROGRAM OUT

Viewers get one seamless live program.

The AI never encodes the final video, so its job stays a decision.

Each camera change is a JSON command over the OBS-WebSocket API.

It builds on OBS Studio and vMix, switchers that already exist.

WHERE IT FITS

SPORTS STREAMING IS THE FIRST TARGET

IN PLAY	Tracks the ball and the players, and holds wide angles through fast action.
PLAY STOPS	Cuts to close-ups, crowd reactions and VIPs in the stands.
ALL GAME	Matches color across cameras and keeps blurry or shaking feeds off the stream.

Validation starts on basketball and soccer footage; events and live shows can reuse the same pipeline.

TECHNICAL RISK

THREE TECHNICAL RISKS SHAPE THE BUILD

RISK	WHY IT IS HARD	DESIGN RESPONSE
COMPUTE 4+ LIVE FEEDS	YOLO, face recognition and color grading on 4+ feeds are GPU-heavy.	One RTX 5090 PC in a flight case, analyzing downscaled copies.
LATENCY UNDER 100 MS	Read, analyze and command must all finish in time for live sports.	The AI sends a command and leaves video encoding to the switcher.
CONTEXT THE GAME'S STORY	Lingering on a player after a missed shot takes more than detection.	Learn from SoccerNet, which marks where human directors cut.

COMPUTE PLAN · PROPOSED

ONE POWERFUL UNIT VALIDATES THE IDEA AND TRAVELS TO THE VENUE

STEP 1 · BASELINE

ONE RTX 5090 PC

A compact PC with a 32 GB RTX 5090 runs the AI and the switcher together.

MEASURE: ms per feed, per model

STEP 2 · OPTIMIZE

CUT THE COST PER FEED

TensorRT builds of YOLOv8, GPU video decoding, and face recognition only when play stops.

TARGET: 4+ feeds under 100 ms

STEP 3 · DEPLOY

FLIGHT-CASE INSTALL

The same PC travels in a flight case: connect power, the camera network and the program output.

ON SITE: power, network, output

Analysis stays on 480p/720p copies; the step 1 benchmark shows whether a second GPU is needed.

VALIDATION

OPEN DATASETS REPLACE CAMERAS FOR TESTING

BASKET-MULTIVIEW

Up to 89 synchronized views of the same basketball plays, in 1080p and 4K.

TESTS: COLOR MATCHING, BEST ANGLE

SOCCERNET

Over 3,000 multi-view videos of fouls and goals, marked where a human director cut.

TESTS: LEARNING HUMAN CUT DECISIONS

TEAMTRACK

Ultra-high-resolution full-pitch views of soccer, basketball and handball.

TESTS: WIDE-ANGLE BALL AND PLAYER TRACKING

MULTISPORTS

Player bounding boxes come pre-calculated, so detection training can be skipped.

TESTS: DIRECTING LOGIC ON ITS OWN

THE VIRTUAL STADIUM

A VIRTUAL STADIUM PROVES THE LOOP END TO END

STEP 1

DOWNLOAD

Take 3 to 4 synchronized angles of a single play.

STEP 2

LOOP

Replay the MP4 files as local NDI or RTSP streams with FFmpeg.

STEP 3

POINT

Aim the vision mixer at the local streams as if they were live.

STEP 4

WATCH

Check in OBS Studio that the AI cuts on the recorded action.

No physical cameras are needed to prove the full loop, from ingest to cut.

ROADMAP · DRAFT

THREE PHASES LEAD FROM SIMULATION TO A LIVE PILOT

PHASE 1 · [DATES]

VIRTUAL STADIUM

Prove ingest, analysis, directing rules and switching on open datasets.

DONE WHEN: AI cuts match the recorded action

PHASE 2 · [DATES]

LIVE CAMERAS

Connect physical cameras and run four or more feeds on GPU hardware.

DONE WHEN: the loop runs under 100 ms

PHASE 3 · [DATES]

PILOT EVENT

Direct a real broadcast for [pilot partner] with no one at the mixer.

DONE WHEN: [pilot success measure]

Phases follow the testing plan in the blueprint; dates and pilot partner are still to be set.

SUCCESS CRITERIA

FOUR CHECKS DECIDE WHETHER THE IDEA WORKS

ON THE ACTION

CUTS FOLLOW THE PLAY

The AI switches camera when the recorded action calls for it, checked in OBS Studio.

UNDER 100 MS

THE LOOP IS FAST ENOUGH

Reading a frame, analyzing it and commanding the switcher fit the live-sports budget.

3–4 SECONDS MINIMUM

SHOTS HOLD STEADY

No cut arrives faster than the minimum shot duration, so the picture never flickers.

BAD FEEDS REJECTED

ONLY CLEAN PICTURES AIR

Blurry, shaking or out-of-focus cameras are never selected for output.